NOHELIA ESTEFHANIA TACCA APAZA

Nuñoa (Melgar, Puno) & Arequipa, Peru | +51 986699911 | ntacca.qh@gmail.com
Academic Portfolio

RESEARCH INTERESTS: Indigenous Language Revitalization (Southern Quechua) • Low-Resource Natural Language Processing (NLP) & Computer-Assisted Language Learning (CALL) • Embodied Cognition & Second Language Acquisition (SLA) • Andean Cultural & Indigenous Studies • Andean Digital Humanities & Indigenous Data Sovereignty

EDUCATION

2025 – 2026 - Visiting International Student and Audit Student (Non-Degree), Cornell University, Ithaca, NY, USA (*Sponsored by the Fulbright Program*, Fall 2025 – Spring 2026)

- Completed graduate and undergraduate coursework in Latin American Studies and Pedagogical Methodologies while serving as primary Quechua language instructor.
- **Graded Coursework:** International Teaching Assistant Development Program Course I (Fall 2025); International Teaching Assistant Development Program Course II (Spring 2026).
- **Audited Coursework (Non-Degree):** Indigenous Issues in Global Perspectives; Exploring Strategies for Teaching and Learning; Intermediate Portuguese for Spanish Speakers I; The First American University.

2020 – 2024 - Bachelor of Science (B.S.) in Systems Engineering, *Universidad Nacional de San Agustín de Arequipa (UNSA)*, Arequipa, Peru

- **Research Focus:** Software architecture, human-centered computing, and educational technologies for indigenous language preservation and social impact.

2018 (Aug – Dec) - International Exchange Student, International Business and Research Seminar Participant, *Universidad Santo Tomás (USTAMED)*, Medellín, Colombia

2017 (Mar – Jul) - International Exchange Student, Public Accounting Program (CRISCOS Scholar), *Universidad Católica del Norte (UCN)*, Antofagasta, Chile

2014 – 2018 - Bachelor of Science (B.S.) in Accounting, *Universidad Nacional de San Agustín de Arequipa (UNSA)*, Arequipa, Peru

FELLOWSHIPS, GRANTS & ACADEMIC HONORS

- **2026 - Aspire Social Impact Seed Fund Award**, *Aspire Institute* - Awarded competitive seed funding to lead the Youth Leadership and Community Empowerment Initiative conducted entirely in Quechua in Santa Rosa (Melgar, Puno, Peru).
- **2026 - SIMBig Shared Task Honorable Mention**, *Universidad Católica San Pablo*, Arequipa, Peru | Shared Task Portal

- **2025 – 2026 - Fulbright Foreign Language Teaching Assistant (FLTA) Fellowship**, *U.S. Department of State & Institute of International Education (IIE)* - Hosted at Cornell University, Ithaca, NY, USA | Fellowship Program
- **2025 – 2026 - Language Resource Center (LRC) Grant (Project Director)**, *Cornell University* - Awarded funding for *"Pachamamawan kawsayninchismanta, Our Life with the Pachamama: Real-Life Audiovisual Resources for Elementary Quechua"* | Audiovisual Repository
- **2025 – 2026 - Language Resource Center (LRC) Grant (Project Collaborator)**, *Cornell University* - Awarded funding with Prof. Vanessa Gubbins for *"Intiq wawankunaq willakuyninkuna: Andean Stories in Elementary Quechua"* | Interactive Flipbook
- **2025 - Aspire Leaders Program Fellow (Cohort 3)**, *Aspire Institute* (Founded by Harvard Business School faculty) | Program Overview
- **2024 – 2025 - Code Your Future Scholarship**, Backend Software Development Training in Java.
- **2023 - International Student Festival in Trondheim (ISFiT) Selected Delegate**, Trondheim, Norway - Selected to represent Peru in global dialogues on educational equity and social impact | Festival Website
- **2022 – 2023 - College Horizons Outreach Program (CHOP+) Fellowship**, *U.S. Embassy in Peru* (Jan 2022 – Oct 2023) - Competitive scholarship for indigenous Quechua youth focused on advanced English, social entrepreneurship, and leadership | Program Video
- **2022 - Grace Hopper Celebration (GHC) Scholar**, *AnitaB.org* - Awarded competitive international fellowship recognizing emerging women leaders in computing and technology.
- **2017 - CRISCOS Academic Mobility Scholarship**, *Consejo de Rectores por la Integración de la Subregión Centro Oeste de Sudamérica* - Full academic exchange grant at Universidad Católica del Norte, Chile.
- **2014 – 2018 - Beca Permanencia (Undergraduate Academic Excellence and Retention Scholarship)**, *PRONABEC – Peruvian Ministry of Education*.

RESEARCH & DIGITAL HUMANITIES PROJECTS

2025 – Present - Lead Researcher, *Embodied Cognition and Andean Choreography in Quechua SLA ("Kuntur Tusuyta Tususunchis")* | Presentation Slides

- Developed an interdisciplinary pedagogical framework repositioning traditional Andean dance not merely as folkloric performance, but as a measurable cognitive technology for Quechua Second Language Acquisition (SLA).
- Integrated Embodied Cognition (Macedonia & von Kriegstein, 2012), Total Physical Response (Cantoni, 1999), and Affective Filter reduction (Krashen, 1982) to map choreographic gestures onto agglutinative Quechua suffixes.

2025 – Present - Co-Investigator, *Paqocha: A Retrieval-Augmented Model for Spanish to Cusco-Collao Quechua Translation*

- Architected an innovative Spanish-to-Quechua (Cusco-Collao variety) machine translation system built on a Retrieval-Augmented Generation (RAG) framework, achieving benchmark scores of 21.24 BLEU and 70.46 chrF while addressing morphosyntactic complexity and indigenous data sovereignty.

2025 – Present - Co-Investigator, *Paqocha Analyzer: Audio Concordance and Morphological Breakdown Platform for Puno Quechua* | Web Platform

- Designed and deployed a reproducible, low-cost YouGlish-style Computer-Assisted Language Learning (CALL) web platform leveraging the spontaneous-speech corpus of Common Voice Puno Quechua (qxp).
- Built an automated data ingestion pipeline filtering 1,222 transcribed audio recordings through a deterministic, rule-based morphological analyzer that decomposes agglutinative words into roots and suffixes via iterative right-to-left stripping alongside word-level forced timestamp alignment.

2025 – 2026 - Project Director, *Pachamamawan kawsayninchismanta, Our Life with the Pachamama: Real-Life Audiovisual Resources for Elementary Quechua* (*Cornell University LRC Grant*, with Prof. Mary K. Redmond) | Video Collection

- Directed the pedagogical design, filming, transcription, and subtitling of open-access audiovisual learning modules situating elementary Southern Quechua grammar within authentic Andean agricultural and cultural contexts.

2025 – 2026 - Project Collaborator, *Intiq wawankunaq willakuyninkuna: Andean Stories in Elementary Quechua* (*Cornell University LRC Grant*, with Prof. Vanessa Gubbins) | Digital Book

- Co-developed an illustrated digital reader adapting traditional Andean narratives into accessible elementary Quechua for university-level SLA learners.

2023 – 2024 - Quechua Linguistic Data Specialist & Transcriber (Freelance), *Sigma AI*, Remote

- Transcribed, segmented, and annotated spontaneous oral corpora in Puno (Collao) Quechua (qxp) to train speech recognition and low-resource AI language models.

2022 – Present - Assistant Researcher, *Center for Research, Technology Transfer, and Software Development (CiTeSoft), UNSA*, Arequipa, Peru (Aug 2022 – Present) | Institute Website

- Conduct multidisciplinary research at the intersection of Systems Engineering, computational linguistics, and human-centered software development to address educational and linguistic inequities in southern Peru.

CONFERENCE PRESENTATIONS

2026 - Tacca Apaza, N. E., and Machaca Condori, E. E. J. (2026). *"Dance as a Vehicle for the Preservation and Revitalization of the Quechua Language: The Case of the Workshop 'Kuntur Tusuyta Tususunchis'."*

- **Conference on Indigenous Languages of Latin America (CILLA) 2026**, Department of Linguistics and LLILAS Benson Latin American Studies and Collections, *The University of Texas at Austin*, Austin, TX (*Accepted and presented*) | Program • Slides
- **Symposium on American Indian Languages (SAIL) 2026**, *The University of Arizona*, Tucson, AZ (*Accepted and presented*) | Conference Website

- **First Glocal Conference on Quechua Languages and Cultures 2026**, Quechua Language and Culture Program at *University of Pennsylvania* and Hamilton Lugar School of Global and International Studies at *Indiana University Bloomington*, Philadelphia, PA (*Accepted and presented*) | Program

2026 - Merma Portocarrero, J. F., **Tacca Apaza, N. E.**, Fernandez, R., and Luque Ccosi, P. A. (2026). *"Paqocha Analyzer: An Audio Concordancer and Deterministic Morphological Analyzer for Puno Quechua."*

- **International Conference on Information Management and Big Data (SIMBig) – Shared Task**, *Universidad Católica San Pablo*, Arequipa, Peru (*Accepted, presented, and awarded Honorable Mention*) | Shared Task Site • SIMBig Website

2025 – 2026 - Merma Portocarrero, J. F., and **Tacca Apaza, N. E.** (2025–2026). *"Paqocha: A Retrieval-Augmented Model for Spanish to Cusco Quechua Translation."*

- **First Glocal Conference on Quechua Languages and Cultures 2026**, *University of Pennsylvania* and *Indiana University Bloomington*, Philadelphia, PA (*Accepted and presented*) | Program
- **Symposium on American Indian Languages (SAIL) 2026**, *The University of Arizona*, Tucson, AZ (*Accepted and presented*) | Conference Website
- **Society for the Study of the Indigenous Languages of the Americas (SSILA) Annual Meeting 2026**, New Orleans, LA (*Accepted and presented*) | Program

TEACHING EXPERIENCE

2025 – 2026 - Quechua Language Instructor (Fulbright FLTA Fellow), *Latin American and Caribbean Studies Program / Department of Romance Studies, Cornell University*, Ithaca, NY, USA (Fall 2025 – Spring 2026) | Program Website

- **Instructed Courses:** Elementary Quechua I (Fall 2025); Elementary Quechua I and Elementary Quechua II (Spring 2026).
- Facilitated university-level Quechua instruction for undergraduate and graduate students, designing original syllabi, digital multimedia resources, and interactive morphosyntactic exercises.
- Led weekly *Quechua Conversation Hours* at the Cornell Language Resource Center and directed semester-long cultural immersion workshops.

2025 – Present - Quechua Language Instructor, *Language Center, Universidad Nacional de San Agustín de Arequipa (UNSA)*, Arequipa, Peru (Feb 2025 – Jul 2025; Sep 2025 – Present)

- Instruct undergraduate, graduate, and community students in Southern Quechua grammar, oral tradition, and Andean epistemology, integrating digital CALL tools to bridge urban academic spaces with indigenous linguistic heritage.

2024 – Present - Founder and Lead Instructor – "AI for Children" (*IA para Niños*) and Educational Robotics Workshops, *Nuñoa District (Melgar, Puno, Peru)* - In collaboration with HOOP Peru and National Central University (NCU, Taiwan)

- Spearheaded the 1st, 2nd, and 3rd editions of the Educational Robotics Workshop and led the *"AI for Children"* initiative in the native Quechua-speaking community of Nuñoa, Puno.

- Designed culturally responsive curricula introducing rural Andean children to Artificial Intelligence, algorithmic logic, and robotics alongside international volunteers from Taiwan, Colombia, the United States, and Peru | Annual Reports: 2023 • 2024 • 2025 • 2026

2022 – 2023 - Head Coach in Coding and Robotics Program, *Helping Overcome Obstacles in Peru (HOOP Peru)*, Arequipa, Peru (May 2022 – Dec 2023) | Organization Website

- Directed coding (Scratch) and educational robotics instruction for underserved children in Arequipa, mentoring volunteer instructors and developing project-based STEM learning modules.

INVITED TALKS & CULTURAL WORKSHOPS

- **2026 (May) - Invited Guest Speaker**, *Quechua Language and Culture Program* (Invited by Prof. Odi Gonzales), *New York University (NYU)*, New York, NY, USA (*In person*).
- **2026 (Apr 28) - Invited Guest Speaker and Workshop Co-Leader**, *"The Living Voice of the Andes: An Interactive Journey into Quechua Culture with Nohelia Tacca and Eddy Machaca"*, *Mount Holyoke College*, South Hadley, MA, USA (*In person*) | Event Page
- **2026 (Apr 09) - Invited Roundtable Panelist**, *Indigenous Language Politics: "Acoustic Colonialism" and Movements for Resurgence* - A conversation with Dr. Luis Cárcamo-Huechante, Dr. María Aguilar, Prof. Nohelia Tacca Apaza, and Dr. Polly Lauer, *Cornell University*, Ithaca, NY, USA (*In person*).
- **2026 (Mar 31) - Invited Workshop Leader**, *"An Immersive and Interactive Exploration of Quechua Language and the Living Traditions of the Andes"*, *SUNY Oneonta*, Oneonta, NY, USA (*In person*).
- **2026 (Feb 27) - Invited Roundtable Panelist**, *Quechua Teaching in Higher Education* (Moderated by Elwin Huaman), held as part of the First Glocal Conference on Quechua Languages and Cultures 2026, *University of Pennsylvania* and *Indiana University Bloomington*, Philadelphia, PA, USA (*Online*) | Event Program
- **2025 (Nov 11) - Invited Panelist**, *"Fulbright Brings the World to Us"*, *SUNY Oneonta (Morris Hall)*, Oneonta, NY, USA (*In person*) | Event Bulletin
- **2025 (Nov 07) - Invited Guest Speaker and Workshop Co-Leader**, *"The Living Voice of the Andes: An Interactive Journey into Quechua Culture with Nohelia Tacca and Eddy Machaca"*, *Mount Holyoke College*, South Hadley, MA, USA (*In person*) | Event Speakers
- **2025 (Oct 25) - Invited Workshop Co-Leader**, *"The Secret of the Andes: Where Do Llamas and Alpacas Come From?"*, *Language Resource Center World Languages Day*, *Cornell University*, Ithaca, NY, USA | Event News
- **2025 – 2026 - Guest Lecturer in Spanish for Heritage Speakers**, *Department of Romance Studies, Cornell University*, Ithaca, NY, USA (Invited by Prof. Mary K. Redmond):
    - *"To Be and Speak Quechua – Ser y hablar Quechua"* (Fall 2025, two sections) | Slides
    - *"Beyond Our Roots – Llaqtanchismanta, más allá de nuestras raíces"* (Spring 2026, two sections, with Prof. Eddy Machaca) | Slides
    - *"My Family: Where We Come From, Where We Are Going – Aylluy: De dónde venimos, hacia dónde vamos"* (with Prof. Eddy Machaca) | Slides
- **2025 – 2026 - Cultural Workshop Leader and Presenter**, *Cornell Quechua Program, Cornell University*, Ithaca, NY, USA:
    - **Lead Organizer:** Andean Dances (*"Tususunchis"*); Harvesting Andean Potatoes (*"Papa Hasp'iy"*); Andean Textiles (*"Punchunchista Sirasunchis"*); Making Our Ponchos (*"Punchu Ruway"*).

- **Co-Leader:** Learning About Andean Music Through the Pan Flute (*"Takisunchis Antarawan"*).

COMMUNITY LEADERSHIP, OUTREACH & PROFESSIONAL EXPERIENCE

2026 - Project Lead, *Aspire Youth Leadership and Community Empowerment Initiative*, Santa Rosa (Melgar, Puno, Peru)

- Led educational empowerment, leadership mentoring, and digital inclusion workshops conducted in Quechua for high-Andean youth in Santa Rosa, Puno, fostering pathways to higher education and community-driven innovation.

2024 – Present - Global Shaper, *World Economic Forum (Arequipa Hub)*, Arequipa, Peru (May 2024 – Present) | Hub Overview

- Co-design and execute high-impact community initiatives focused on educational equity (SDG 4), youth leadership, and decentralized technological access in southern Peru.

2024 – Present - Founder and Lead Instructor, *"Discovering Opportunities" (Descubriendo Oportunidades)*, Nuñoa District (Melgar, Puno, Peru) - In collaboration with National Central University (NCU, Taiwan)

- Organized and led a mentorship speaker series connecting rural high school students in Puno with international higher education scholarships and post-secondary pathways.

2024 – 2025 - Organizer, *Workshop Series "Empowering Andean Youth Through Social Leadership"*, Melgar and Azángaro Provinces, Puno, Peru

- Delivered youth leadership workshops at Domingo Savio High School (Nuñoa), Túpac Amaru High School (Nuñoa), Siervos de Dios High School (Crucero), and San Agustín High School (Samán).

2024 – 2025 - Software Development Pre-Professional Intern, *Darshana.io*, Online (Aug 2024 – Apr 2025

LANGUAGES

- Quechua (Native Speaker - Southern / Cusco-Collao and Puno variety, qxp)
- Spanish (Native Speaker)
- English (C1)
- Portuguese (A2)
- French (A1)